TECHNOLOGY STACK - COURIER MANAGEMENT SYSTEM

Platform

- Python Flask / Java Spring Boot
- MySQL Database
- XAMPP Local Server

Data Model

- Tables: Users, Shipments, Tracking_History
- Custom Fields & Relationships
- Validation & 3NF

App & UI

- Bootstrap 5 Responsive UI
- Courier Management App
- Tracking & Booking Pages

Security

- Password Hashing (bcrypt)
- OTP Verification
- Role Based Access (Admin/Agent)

Automation

- Auto Tracking ID: CMS123456
- Auto Cost Calculation
- SMS/Email Alerts via API

Reporting

- Daily Shipments Report
- Admin Dashboard
- Invoice PDF Generation

COURIER MANAGEMENT - DEVELOPMENT MODULES

Implemented Features (Courier Project)

BOOKING MODULE (Code: 40%)

- Flask Route: /book
- Weight -> Cost Calc
- Tracking ID: CMS + random
- MySQL Insert: Shipments table
- Validation: Phone, Pincode

TRACKING MODULE (Code: 30%)

- API: /track/<id>
- Status Timeline UI
- History Table Update
- Agent Location (Mock)
- SMS Alert Integration

ADMIN & AGENT MODULE (Code: 30%)

- Admin: View All, Assign Agent
- Agent: Update Status, POD
- Photo/Signature Upload
- Invoice PDF Generation
- Dashboard Analytics

Development Tools Used:

- Frontend: HTML5, CSS3, Bootstrap 5, JavaScript | Backend: Python Flask (app.py) | Database: MySQL (phpMyAdmin) | Version Control: Git & GitHub
- Testing: Postman for APIs, Manual Testing for Booking Flow | Deployment: Localhost (XAMPP) -> Ready for Cloud
- Lines of Code: ~1500 | Database Tables: 5 | API Endpoints: 8 | Completed: 100% Modules